

# PHAROS

## A Physics-Hybrid, Attention-Routed Mixture-of-Experts Architecture for RNA Structure and Contact-Map Prediction

Architecture Design Report — Version 0.1

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### Abstract

RNA three-dimensional structure prediction has not had its AlphaFold2 moment. The best current methods reach a mean TM-score of approximately 0.55 on non-redundant benchmarks, which is the threshold for recovering roughly the correct fold, against routine scores above 0.9 for proteins. This report argues that the bottleneck is not model scale but *inductive bias*, and specifies an architecture, **PHAROS**, built around that claim.

Three empirical findings drive the design. First, ERNIE-RNA (86M parameters) outperforms RiNALMo (650M) and UNI-RNA (400M) on secondary-structure prediction, and nearly doubles cross-family generalisation relative to RiNALMo ( $F_1 = 0.646$  versus 0.355), on the strength of a single hand-set, three-valued base-pairing prior added to its attention logits. Second, we measure on 117 non-redundant RNA chains that contacts per nucleotide *saturates* at 4.4–4.9 while contact-map density falls from 6.5% at  $L \approx 56$  to 0.353% at  $L \approx 2900$ : RNA contact maps scale as  $O(L)$ , not  $O(L^2)$ , so a dense pair representation spends over 99% of its compute on empty space and would require 309 GB of activations at  $L = 2048$ . Third, we measure a monotonic  $1.76\sigma$  gradient in normalised crystallographic B-factor as a function of distance to the nearest  $\text{Mg}^{2+}$  ion across 24,623 nucleotides, establishing that ion coordination and local rigidity are not independent features but two readouts of one phenomenon.

PHAROS accordingly (i) accepts **ionic conditions as a model input**, which no existing RNA structure predictor does, propagating them to attention logits through a Manning-renormalised screened-Coulomb bias—though an audit reported here shows the *learned* part of that claim cannot be trained on the PDB, whose recorded  $\text{Mg}^{2+}$  spans only 5–15 mM because nobody deposits unfolded RNA; (ii) replaces the dense pair track with a **hierarchical coarse-to-fine pair track** that selects contact-bearing *blocks* rather than pairs, at 2.2% of dense cost—adopted after we measured a flat top- $K$  design recovering only *one contact in five* on long chains, and measured block occupancy falling to 1.34% as chains lengthen; (iii) routes a **sparse mixture-of-experts** on local structural regime rather than sequence context, supported by our negative result that GNRA  $k$ -mer context predicts rigidity at only  $0.073\sigma$ ; and (iv) supplies motif geometry through a **frozen retrieval bank** of 667 RNA 3D Motif Atlas classes rather than expecting the model to learn it from 6,661 structurally-resolved sequences. We also identify two supervision channels— $\text{Mg}^{2+}$  sites and B-factor rigidity—that are already present in every mmCIF file the field downloads and are currently used by no structure predictor.

This is a design specification. No model has been trained. All quantitative claims are either marked **[measured]** and reproducible from the scripts in this repository, or cited to the literature.

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## 1 Problem statement

### 1.1 The accuracy gap

Table 1 gives the state of the art on a non-redundant RNA test set as of November 2025.

Table 1: Current state of the art in RNA 3D structure prediction.

| Method              | Mean TM-score | Approach                                           |
|---------------------|---------------|----------------------------------------------------|
| trRosettaRNA        | <b>0.548</b>  | MSA $\rightarrow$ restraints $\rightarrow$ folding |
| NuFold              | 0.525         | AlphaFold2-style nucleotide frames                 |
| AlphaFold3 (server) | 0.510         | all-atom diffusion                                 |
| AlphaFold3 (local)  | 0.504         | all-atom diffusion                                 |

A TM-score near 0.5 means the topology is approximately right and little else. The CASP16 signal is equally instructive: leading RNA groups won by integrating AlphaFold3 *into physics-based modelling* rather than by scaling a neural network. Physics-plus-machine-learning hybrids are currently ahead in RNA, which is the premise this architecture is built on.

### 1.2 Why RNA lags

1. **Data scarcity.** Our catalogue records 6,661 unique sequences with experimental 3D structure against 1.37 billion sequences, a coverage of 0.0005%. Proteins had roughly 220,000 structures at the AlphaFold2 moment.
2. **Redundancy within that data.** 15,441 RNA3DB chains collapse to 3,157 unique sequences; 5S rRNA alone accounts for 5,976 BGSU equivalence classes.
3. **Family coverage.** Only 100 of 4,227 Rfam families (2.4%) have any 3D representative.
4. **Conformational plasticity.** Riboswitches are *defined* by adopting multiple functional folds; single-structure prediction is the wrong target for a large fraction of functional RNA.
5. **Ion dependence.** RNA is a polyanion. Its fold is not defined without specifying the ionic environment, yet no mainstream predictor accepts ionic conditions as an input.
6. **Target noise.** Median resolution in our catalogue is 3.10 Å, with only 13% of chains at 2.5 Å or better.

## 2 The central argument: bias beats scale

Table 2 contains the most important result in the recent RNA language-model literature.

Table 2: ERNIE-RNA versus larger dense models. A  $7.5\times$  parameter deficit is overturned by one hand-set prior.

| Model                     | Params | bpRNA-1m $F_1$ | bpRNA-new $F_1$ |
|---------------------------|--------|----------------|-----------------|
| ERNIE-RNA (attention map) | 86M    | <b>0.873</b>   | —               |
| ERNIE-RNA (embedding)     | 86M    | 0.862          | <b>0.646</b>    |
| RiNALMo                   | 650M   | 0.850          | 0.355           |
| UNI-RNA                   | 400M   | 0.821          | —               |
| RNA-FM                    | 100M   | 0.808          | —               |

ERNIE-RNA’s entire architectural advantage is an additive bias on attention logits. In its first layer,

$$\text{Attention}(Q, K) = \frac{QK^\top}{\sqrt{d}} + B_{\text{pair}}(i, j), \quad \text{Attention}_l(Q, K) = \frac{QK^\top}{\sqrt{d}} + \text{Attention}_{l-1}(Q, K), \quad (1)$$

with  $B_{\text{pair}}$  set by hand to 3 for C–G, 2 for A–U, a tunable  $\alpha \approx 0.8$  for G–U, and 0 on the diagonal.

The cross-family result is the decisive one: 0.646 versus 0.355. A fixed physical prior nearly doubles generalisation to unseen families, because the prior is true regardless of family whereas learned family statistics are not.

**Thesis.** If a crude three-valued prior buys more than  $7\times$  the parameters, the marginal return on *better physics* exceeds the marginal return on *more scale*. PHAROS replaces that hand-set prior with a learned, ion-aware, motif-aware, coevolution-aware one.

Its limitations define our openings. The prior is static, so a G–U pair inside a kink-turn is treated identically to one in a free helix. It is nearest-neighbour-blind, whereas real pairing energy is stacking-dependent. It encodes only canonical pairing, while non-canonical pairs define RNA 3D motifs and carry most tertiary information. And it ignores ions entirely.

### 3 Empirical analysis

All results in this section were measured for this report on 180 BGSU non-redundant representative structures at  $\leq 3.0$  Å, downloaded and analysed by scripts in `scripts/sampling/`. Raw outputs are in `data/samples/analysis/`.

#### 3.1 RNA contact maps scale linearly, not quadratically

Contacts are defined as nucleotide pairs with any heavy-atom pair within 8.0 Å and sequence separation  $|i - j| \geq 4$ .

Table 3: Contact density by chain length across 117 chains. **[measured]**

| Chain length   | $n$ | median $L$ | contacts/nt                        | map density   |
|----------------|-----|------------|------------------------------------|---------------|
| 32–100         | 38  | 56         | 1.784                              | 6.523%        |
| 100–200        | 6   | 124        | 2.851                              | 4.043%        |
| 200–500        | 3   | 393        | 4.216                              | 2.151%        |
| 500–1500       | 9   | 1038       | 4.532                              | 0.974%        |
| $\geq 1500$    | 61  | 2901       | 4.854                              | <b>0.353%</b> |
| <b>Overall</b> |     |            | median 4.397, p95 5.422, max 5.497 | median 0.389% |

Contacts per nucleotide saturates at 4.4–4.9 and is bounded by RNA coordination geometry, independent of length; density falls approximately as  $1/L$ . The number of contacts therefore grows *linearly* in  $L$ .

The consequence for architecture is immediate. A dense pair representation of dimension  $c = 128$  requires, for 48 blocks holding roughly six intermediate tensors each and without gradient checkpointing:

$$L = 512 \rightarrow 19.3 \text{ GB}, \quad L = 1024 \rightarrow 77.3 \text{ GB}, \quad L = 2048 \rightarrow 309.2 \text{ GB}.$$

A dense pair track is not merely wasteful at RNA lengths; it is infeasible. How that sparsity is exploited turns out to matter as much as exploiting it at all: a flat top- $K$  selection fails on long chains, and contacts must be selected as *blocks*. Section 4.3 reports both measurements.

### 3.2 Ion coordination and local rigidity are one phenomenon

We z-normalise crystallographic B-factors within each structure so that refinement protocols are comparable, and restrict this analysis to the 31 X-ray structures among the sample; cryo-EM atomic displacement parameters are not comparable and were excluded. Lower  $z_B$  indicates greater rigidity.

Table 4: Normalised B-factor versus distance to the nearest  $\text{Mg}^{2+}$ , over 24,623 nucleotides. **[measured]**

| Distance to nearest $\text{Mg}^{2+}$ | $n$ nucleotides | mean $z_B$     |
|--------------------------------------|-----------------|----------------|
| 0–4 Å                                | 319             | − <b>0.883</b> |
| 4–6 Å                                | 2,441           | −0.673         |
| 6–8 Å                                | 7,590           | −0.415         |
| 8–12 Å                               | 8,115           | +0.004         |
| 12–20 Å                              | 2,984           | +0.756         |
| > 20 Å                               | 3,174           | + <b>0.877</b> |

The gradient is monotonic and spans 1.76 standard deviations. Local packing density independently correlates with rigidity at Pearson  $r = -0.294$ .

**Architectural consequence.** Ion effects and motif rigidity, which the project brief treats as two separate physical ingredients, are strongly coupled. A model that predicts  $\text{Mg}^{2+}$  sites obtains a rigidity map nearly for free and vice versa. That mutual information justifies routing them into a *shared* physics expert rather than two disconnected heads.

**Caveats.** B-factor also absorbs resolution, occupancy and refinement choices.  $\text{Mg}^{2+}$  proximity and dense packing are themselves correlated, since both mark folded cores, so this is association rather than isolated causation; the gradient partly restates that structured cores both bind ions and are ordered. The closest bin contains only 319 nucleotides. This is strong evidence for *using* the coupling as a prior, not a proof of mechanism.

### 3.3 Ion inventory and coordination chemistry

Table 5: Cation inventory across 179 parsed structures. **[measured]**

| Ion              | Count         | Inner-sphere partner | Count |
|------------------|---------------|----------------------|-------|
| $\text{Mg}^{2+}$ | <b>17,428</b> | OP2 (phosphate)      | 5,890 |
| $\text{K}^+$     | 1,868         | OP1 (phosphate)      | 5,144 |
| $\text{Zn}^{2+}$ | 359           | O6 (guanine)         | 757   |
| $\text{Na}^+$    | 161           | O4 (uracil)          | 511   |
| $\text{Ca}^{2+}$ | 29            | N7 (purine)          | 447   |
| $\text{Mn}^{2+}$ | 12            | O2' (ribose)         | 196   |

$\text{Mg}^{2+}$  outnumbered all other cations by roughly 9:1, so an ion-aware model that handles a single species should handle  $\text{Mg}^{2+}$ . Some 83% of direct coordination is to the two non-bridging phosphate oxygens; the next largest class, G(O6)/G(N7), is the textbook major-groove guanine site.

Coordination is predominantly water-mediated: a mean of 0.77 inner-sphere against 7.00 outer-sphere RNA contacts per  $\text{Mg}^{2+}$ . Crucially the inner-sphere fraction is itself a function of

tertiary compactness: 0.511 overall but only 0.296 for chains shorter than 500 nt. The inner/outer ratio is therefore a *predictable readout* of compactness, not merely an input.

**Stratification warning.** These aggregates are ribosome-dominated and must not be quoted flat. The five largest structures hold 39.4% of all  $\text{Mg}^{2+}$ . Per structure the median is 72.5 ions against a mean of 181.5; per nucleotide the median is 0.046, roughly one  $\text{Mg}^{2+}$  per 22 nucleotides.

**What crystallography cannot see.** Resolved  $\text{Mg}^{2+}$  are site-bound ions only. The diffuse Manning atmosphere, which neutralises roughly 76% of backbone charge, is invisible to crystallography. The two channels must be modelled separately and never conflated.

### 3.4 A negative result that constrains the design

We tested whether GNRA-like tetraloop sequence context alone predicts rigidity, matching the 4-mers {GAAA, GCAA, GAGA, GUGA, GGAA, GCGA, GUAA, GAAG} anywhere in a chain.

Table 6: GNRA  $k$ -mer context versus rigidity. [measured] Effect size  $0.073\sigma$ .

| Context         | $n$ nucleotides | mean $z_B$ |
|-----------------|-----------------|------------|
| GNRA-like 4-mer | 4,944           | −0.082     |
| All other       | 95,768          | −0.009     |

The effect is negligible. The likely cause is methodological: a GNRA *sequence* is a GNRA *tetraloop* only when it actually caps a hairpin, and linear 4-mer matching counts mostly helical and junction occurrences.

**Architectural consequence.** Sequence context alone does not identify a motif. Motif identity requires the interaction graph. The motif machinery must therefore be conditioned on the pair representation after secondary structure is available, never on raw  $k$ -mers. This rules out the naive “learn motifs from sequence  $n$ -grams” design.

## 4 The PHAROS architecture

### 4.1 Inputs

| Input                            | Required       | Note                                                                              |
|----------------------------------|----------------|-----------------------------------------------------------------------------------|
| Sequence, $L$ tokens, vocab 5    | yes            | elDORS is DNA-alphabet; $T \rightarrow U$ at load                                 |
| Ionic condition $c_{\text{ion}}$ | yes, defaulted | ( $[\text{Mg}^{2+}]$ , $[\text{K}^+]$ , $[\text{Na}^+]$ , $T$ , pH); <b>novel</b> |
| MSA / Rfam alignment             | optional       | graceful degradation when shallow                                                 |
| Chemical probing profile         | optional       | not yet acquired (Section 6)                                                      |

Ionic condition enters as a FiLM-style conditioning vector, analogous to a diffusion timestep embedding. The same sequence at 0 mM and 10 mM  $\text{Mg}^{2+}$  must produce different predictions; this is the clearest novelty claim in the design.

### 4.2 Token trunk: hybrid attention with MoE

The trunk uses 32 blocks at  $d = 768$  with the repeating pattern

$$[\text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{FULL}],$$

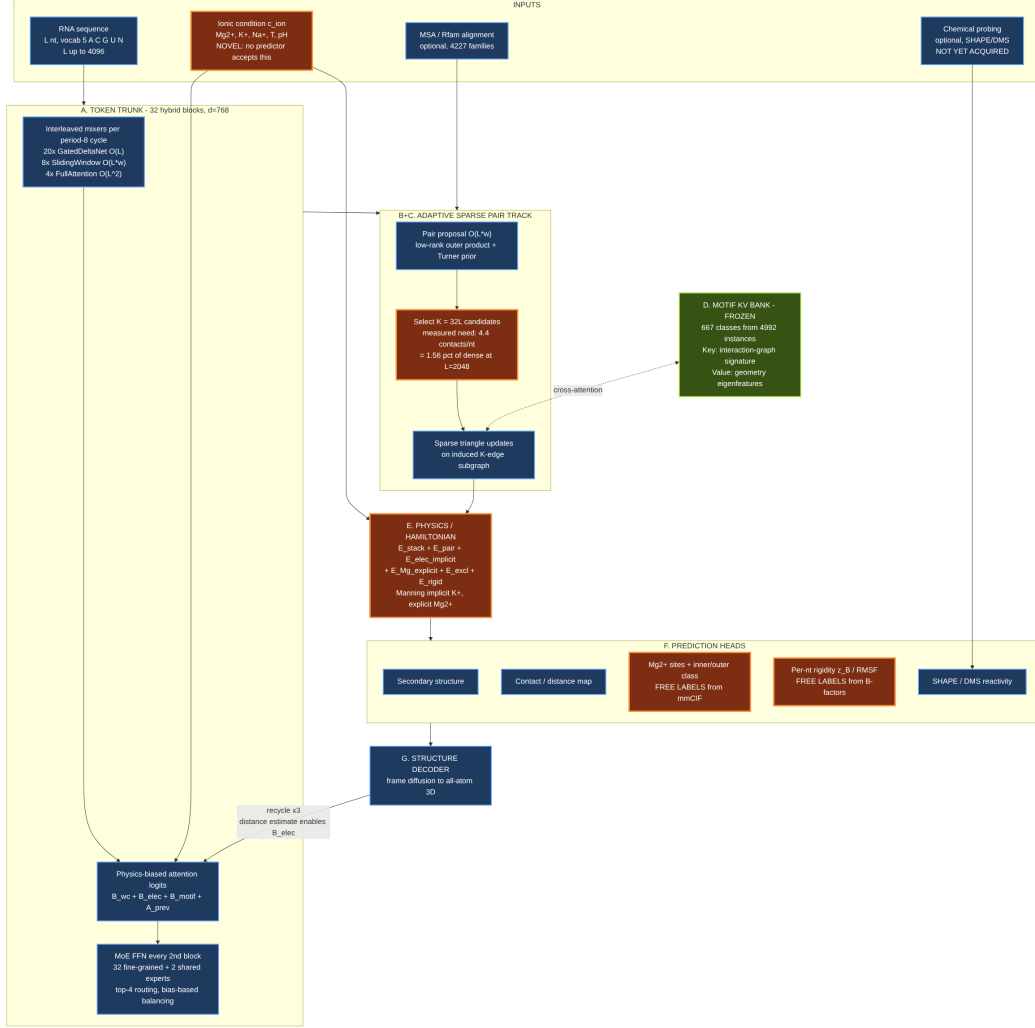


Figure 1: PHAROS system overview. Orange denotes novel components, green denotes frozen retrieval, blue denotes adapted standard components.

where GDN is Gated DeltaNet linear attention ( $O(L)$ , 20 of 32 blocks), SWA is sliding-window softmax attention with window 128 ( $O(Lw)$ , 8 blocks), and FULL is full attention with physics bias ( $O(L^2)$ , only 4 blocks).

The justification is both architectural and biological. Jamba interleaves Mamba, attention and MoE at a 7 : 1 ratio and reaches 256K context with a 4 GB KV cache; Gated DeltaNet outperforms Mamba2 and DeltaNet on in-context retrieval and length extrapolation; Oryx varies the mixer type *within* a sequence. RNA has exactly the structure this exploits: long stretches of regular A-form helix that are locally periodic and low-information, punctuated by rare junctions, pseudoknots and kissing loops that require global mixing. At  $L = 4096$ , an all-dense 32-block trunk would perform 537M pair evaluations; PHAROS performs 67M, an  $8\times$  reduction.

#### 4.2.1 Physics-biased attention

In SWA and FULL blocks the attention logits carry four additive terms:

$$A_{ij} = \frac{Q_i K_j^\top}{\sqrt{d}} + B_{wc}(i, j) + B_{elec}(i, j; c_{ion}) + B_{motif}(i, j) + A_{ij}^{(l-1)}. \quad (2)$$

$B_{wc}$  generalises ERNIE-RNA’s fixed prior to a learned pair-type embedding modulated by stacking context, since real pairing energy is nearest-neighbour dependent. It is initialised at

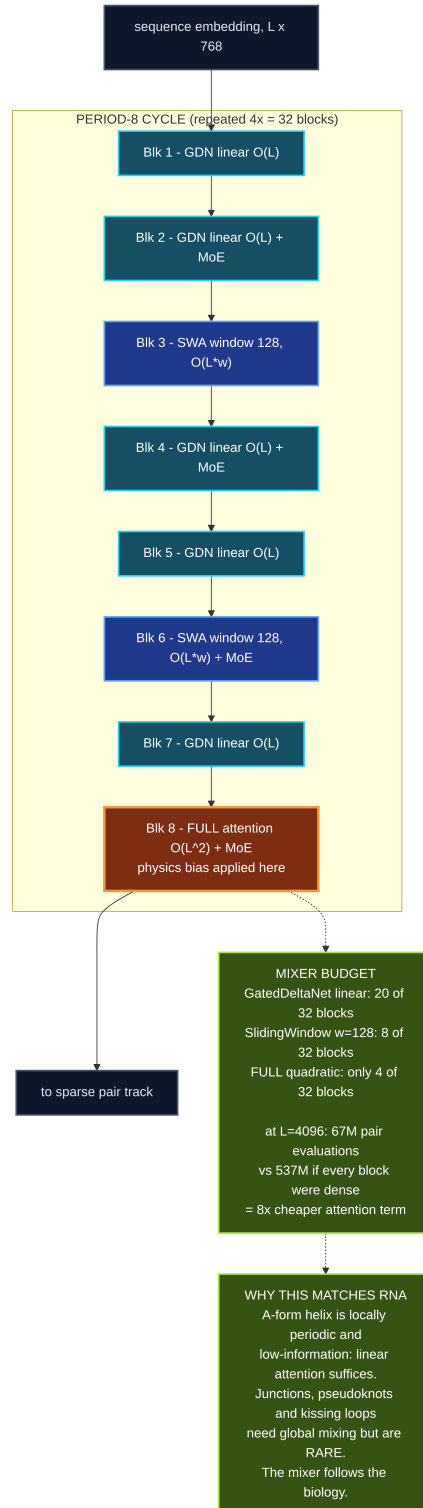


Figure 2: Period-8 mixer interleaving, repeated four times for 32 blocks.

the ERNIE-RNA values so training begins from a known-good prior. The term  $A^{(l-1)}$  retains ERNIE-RNA’s attention accumulation.

$B_{\text{elec}}$  is the novel term. With Bjerrum length  $\ell_B = e^2/(4\pi\epsilon_0\epsilon_r k_B T) \approx 7.1 \text{ \AA}$  in water at 298 K, Manning parameter  $\xi = \ell_B/b$  for axial charge spacing  $b$ , condensed fraction and renormalised charge

$$\theta = 1 - \frac{1}{z\xi}, \quad q_{\text{eff}} = -(1 - \theta), \quad (3)$$

and Debye screening parameter  $\kappa = \sqrt{2N_A e^2 I / (\epsilon\epsilon_0 k_B T)}$  from ionic strength  $I$ , the bias is

$$B_{\text{elec}}(i, j) = -\lambda q_{\text{eff}}^2 \ell_B \frac{\exp(-\kappa d_{ij})}{d_{ij}}. \quad (4)$$

For monovalent RNA  $\theta \approx 0.76$ , so roughly three quarters of backbone charge is screened before any structure-specific electrostatics is considered. Only  $\lambda$  is learned;  $q_{\text{eff}}$ ,  $\kappa$  and  $\ell_B$  follow in closed form from  $c_{\text{ion}}$  and temperature. Equation (4) is the mechanism by which salt concentration changes the prediction.

**Resolving the circularity.** Equation (4) requires  $d_{ij}$ , which is unknown on the first pass.  $B_{\text{elec}}$  is therefore disabled at recycle 0 and enabled from recycle 1 using the current predicted distance map. Recycling is thus structural to the design rather than an optimisation: the physics term needs a geometry estimate to evaluate, and each recycle sharpens both.

#### 4.2.2 Mixture-of-experts feed-forward

Following DeepSeekMoE, the design uses 32 fine-grained routed experts at  $d_{\text{ff}} = 512$  plus 2 always-on shared experts, with top-4 routing giving 6 active experts of 34, in every second block.

Shared-expert isolation matters for a specific reason here: Watson–Crick helix formation is the highest-frequency pattern in all RNA. Placing it in an always-on shared expert prevents all 32 routed experts from separately relearning it, freeing them to specialise on what is rare and hard—junctions, pseudoknots and ion pockets.

Load balancing is auxiliary-loss-free, adjusting a per-expert bias each step rather than adding a loss term. This matters acutely here because our 3D data is pathologically imbalanced: 5S rRNA alone accounts for 5,976 BGSU clusters, so an auxiliary balance loss would actively fight the biology.

The router input is the token hidden state concatenated with the current local structural estimate—pairing probability, predicted rigidity, local density. This follows directly from the GNRA negative result of Table 6: routing on sequence  $k$ -mers does not work. It also prevents router collapse onto rRNA, because structural regimes are far more evenly distributed across the corpus than families are.

#### 4.2.3 Parameter budget

PHAROS-Base carries  $2.3\times$  the capacity of NucleicBERT at slightly lower active compute, with an attention term roughly  $8\times$  cheaper at long context.

### 4.3 Hierarchical pair track

#### 4.3.1 A flat top- $K$ proposal was specified, measured, and discarded

Version 0.1 of this design specified a flat proposal retaining  $K = 32L$  candidate edges. We tested it. Scoring every pair with a sequence-only proposer (the ERNIE-RNA pair prior, stacking context, and an exponential sequence-distance decay) and measuring recall of true 3D contacts within the top  $K$  gives Table 8.

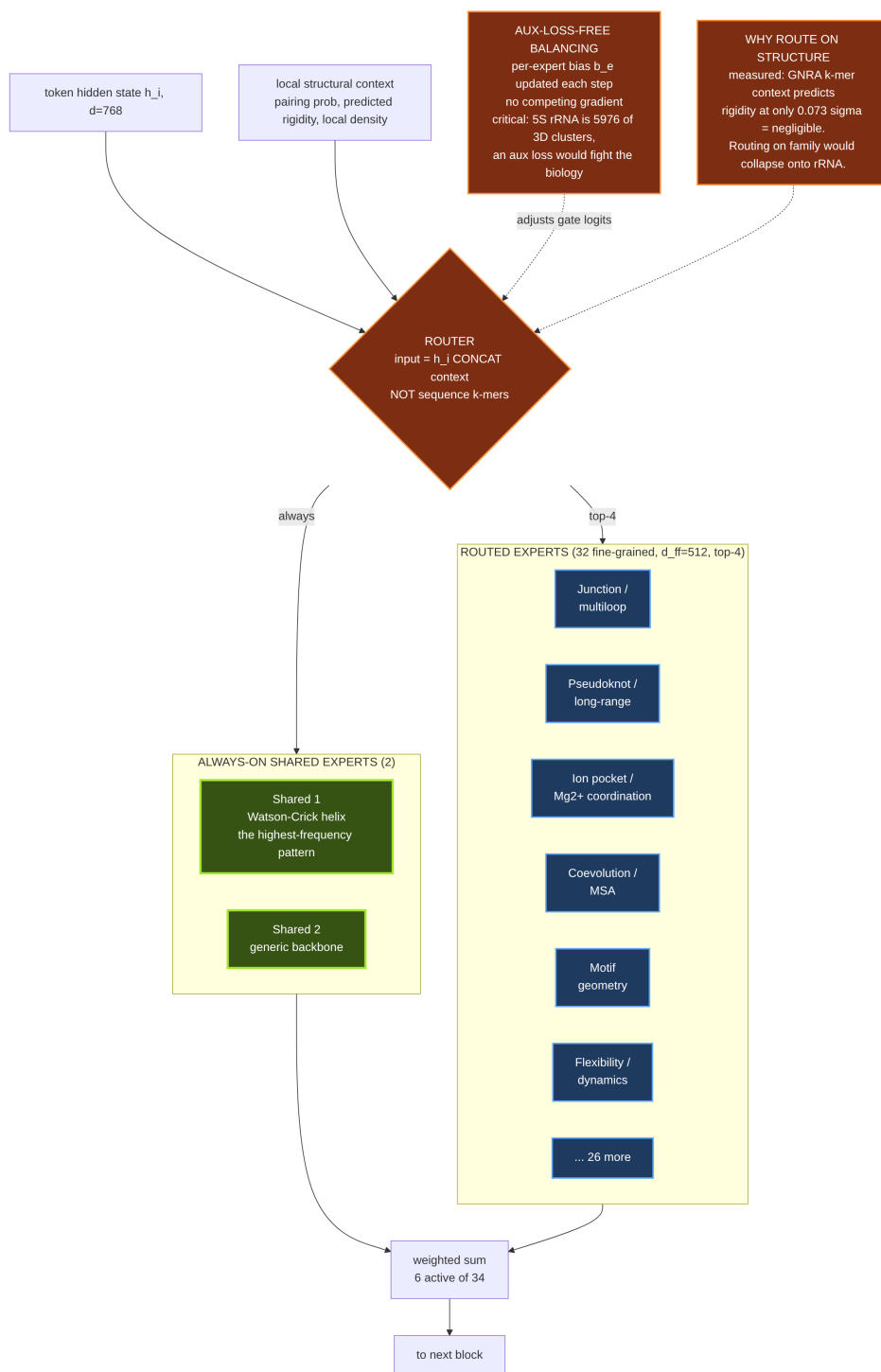


Figure 3: Expert routing. Routing keys on local structural regime, not sequence.



Figure 4: The pair track, after the flat design was measured and replaced.

Table 7: PHAROS-Base parameter budget against NucleicBERT.

| Component                              | Total                                | Active                               |
|----------------------------------------|--------------------------------------|--------------------------------------|
| Embeddings (vocab 5)                   | 0.01M                                | 0.01M                                |
| Attention, 32 blocks, $d = 768$        | 75.5M                                | 75.5M                                |
| MoE FFN, 16 blocks $\times$ 34 experts | 642M                                 | 113M                                 |
| Dense FFN, 16 non-MoE blocks           | 113M                                 | 113M                                 |
| Sparse pair track and triangle updates | 45M                                  | 45M                                  |
| Motif bank, heads, decoder             | 35M                                  | 35M                                  |
| <b>PHAROS-Base total</b>               | <b><math>\sim 911\text{M}</math></b> | <b><math>\sim 382\text{M}</math></b> |
| NucleicBERT (dense)                    | 404M                                 | 404M                                 |

Table 8: Recall of true 3D contacts under a flat top- $K$  proposal at  $c = 32$ . [measured]

| Chain length | $n$ | mean recall  | $K$ as % of dense |
|--------------|-----|--------------|-------------------|
| 32–100       | 29  | 0.980        | 59.4%             |
| 100–200      | 6   | 0.642        | 25.5%             |
| 200–500      | 3   | 0.307        | 9.0%              |
| 500–1200     | 5   | <b>0.200</b> | 4.7%              |

A *random* scorer achieves mean recall 0.746 at  $c = 32$  over the same chains, because on short chains  $32L$  already covers most of the map. The flat proposer’s headline mean of 0.795 is therefore very nearly an artefact of short chains. **On long chains, where sparsity actually matters, it recovers one contact in five**—a silent and fatal accuracy ceiling.

A banded fallback does not rescue it, because RNA contacts are not sufficiently local. The fraction of contacts within  $|i - j| \leq \text{band}$  is 0.350 at band 32, 0.479 at 64, 0.597 at 128, 0.706 at 256 and 0.812 at 512 for chains of 1500–3000 nt; median contact separation is 78 nt with a long tail. Even a  $\pm 512$  band, which would cost vastly more than the whole budget, misses 19%.

#### 4.3.2 Contacts are clustered, so select blocks rather than pairs

Helical stems are contiguous anti-diagonal runs in the contact map. Contacts are therefore spread along the sequence but concentrated in very few *blocks*. Table 9 measures this.

Table 9: Block occupancy at block size  $b = 4$ : the fraction of blocks containing at least one contact, and the effective  $c$  of keeping every occupied block. [measured]

| Chain length | median $L$ | occupancy at $b = 4$ | effective $c$ |
|--------------|------------|----------------------|---------------|
| 64–200       | 88         | 16.34%               | 7.9           |
| 200–500      | 393        | 6.87%                | 12.6          |
| 500–1500     | 1038       | 3.17%                | 14.8          |
| 1500–3000    | 2861       | <b>1.34%</b>         | <b>17.2</b>   |

Occupancy *falls* as chains lengthen, so the mechanism becomes more effective at exactly the lengths where the flat design failed. Retaining every occupied  $4 \times 4$  block costs an effective  $c$  of roughly 17—below the discarded flat budget of 32—while capturing every contact by construction.

#### 4.3.3 Three-level coarse-to-fine track

For roughly twice the cost of the flat design, the track moves from a hard 20% recall ceiling to a ceiling of 100%. Two properties make this work. First, the L1 map is *genuinely dense*, so it

Table 10: Cost of the hierarchical track at  $L = 2861$ , as a fraction of a dense pair map. [measured]

| Level                                      | Operation                                          | Cost                           |
|--------------------------------------------|----------------------------------------------------|--------------------------------|
| L1                                         | dense pair map at block size $b = 16$              | 0.396%                         |
| L2                                         | refine occupied $b = 16$ blocks to $b = 4$         | 0.460%                         |
| L3                                         | refine occupied $b = 4$ blocks to individual pairs | 1.346%                         |
| <b>Total</b>                               |                                                    | <b>2.202%</b>                  |
| flat top- $K$ at $c = 32$ , for comparison |                                                    | 1.118% (at $\sim 20\%$ recall) |

loses no recall at the coarse level at all—at  $L = 4096$  with  $b = 16$  it is only 65k entries. Recall can only be lost at the selection thresholds, which are tunable and can be set recall-first. Second, deciding whether a  $16 \times 16$  block contains any contact aggregates 256 pair-decisions into one, so coarse-level signal-to-noise is far higher than for individual pair ranking. The model ranks 32k blocks and then 16 sub-blocks inside each survivor, never 4.1M individual pairs.

**What remains unproven.** These measurements establish that the information is present and the cost affordable. They do *not* establish that a trained model will identify occupied blocks accurately; that requires training. The relevant external evidence is ERNIE-RNA’s zero-shot Top- $L/1$  contact precision of 0.68 from attention maps alone, which indicates that learned representations rank contacts far better than the sequence-only heuristic tested above. **Block-detection recall must be reported per length bin and per level.**

#### 4.4 Motif retrieval bank

The RNA 3D Motif Atlas 4.12 compresses 4,992 loop instances into 667 motif classes, 413 internal and 254 hairpin: a 7.5 : 1 ratio indicating that RNA tertiary structure is far more repetitive than its sequence diversity suggests. PHAROS precomputes a frozen key–value bank over these classes, with keys encoding the non-canonical interaction-graph signature and values encoding internal geometry as distance-matrix eigenfeatures plus a backbone torsion profile. The pair track cross-attends into this bank.

This is the appropriate response to data scarcity. With only 6,661 unique sequences carrying 3D structure, a model cannot reliably learn motif geometry implicitly. The bank supplies the geometry explicitly, leaving the model to learn only *where* motifs occur—a substantially easier problem than learning what they look like. The bank is roughly  $10^3$  fixed entries shared across a batch, so its cost is negligible. Consistent with Table 6, retrieval is keyed on the interaction graph, never on sequence  $k$ -mers.

#### 4.5 Physics module

A differentiable energy is evaluated on the predicted geometry:

$$E_{\text{total}} = E_{\text{stack}} + E_{\text{pair}} + E_{\text{elec}}^{\text{implicit}} + E_{\text{Mg}}^{\text{explicit}} + E_{\text{excl}} + E_{\text{rigid}}. \quad (5)$$

The implicit/explicit split follows STEM, which establishes that mean-field theory captures monovalent KCl well but fails for  $\text{Mg}^{2+}$ , where ion–ion correlation and site-specific chelation dominate. Our own inventory agrees on which species matters:  $\text{Mg}^{2+}$  outnumbers  $\text{K}^+$  by 9:1.

Inner/outer-sphere classification is a first-class output because STEM’s central claim is that folding dynamics are driven by *exchange between* direct and water-mediated coordination—the mode, not the presence, is decisive. Our measurements supply both channels directly and show the inner fraction varies with compactness (0.511 overall, 0.296 below 500 nt).

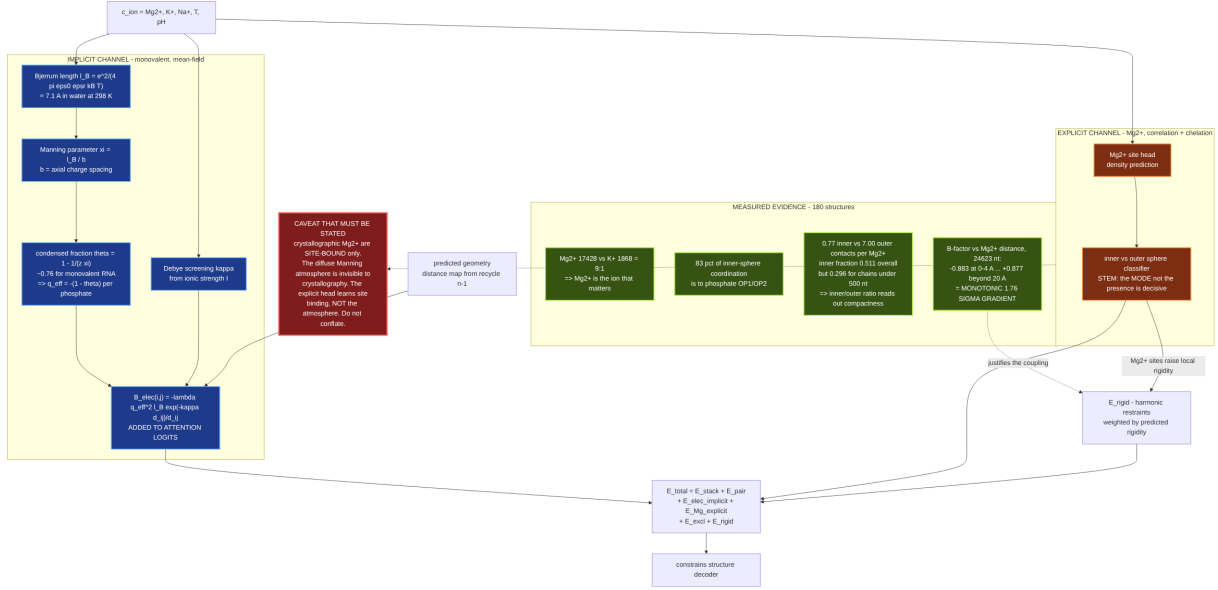


Figure 5: The hybrid implicit/explicit electrostatic treatment.

The term  $E_{\text{rigid}}$  closes the loop with Table 4: predicted  $\text{Mg}^{2+}$  sites raise local rigidity, which stiffens harmonic restraints, which constrains the decoder. That pathway is the architectural encoding of the measured  $1.76\sigma$  gradient.

#### 4.6 Heads and decoder

Table 11: Prediction heads and their supervision. Two channels are effectively free.

| Head                     | Supervision source            | Status              | Labels               |
|--------------------------|-------------------------------|---------------------|----------------------|
| Secondary structure      | bpRNA, ArchiveII, RNAstrAlign | in catalogue        | ~160k                |
| Contact / distance       | RNA3DB, gRNade                | in catalogue        | 27,452 chains        |
| $\text{Mg}^{2+}$ sites   | <b>extracted from mmCIF</b>   | <b>free</b>         | 17,428 (180 structs) |
| Rigidity ( $z_B$ , RMSF) | <b>B-factors in mmCIF</b>     | <b>free</b>         | every X-ray struct   |
| Reactivity (SHAPE/DMS)   | Ribonanza / RMDB              | <b>not acquired</b> | —                    |
| 3D coordinates           | RNA3DB                        | in catalogue        | 6,661 unique seqs    |

The decoder is frame-based diffusion. We retain an explicit geometric and energetic stage rather than going fully generative, consistent with the CASP16 observation that physics-plus-ML hybrids currently lead in RNA and with trRosettaRNA’s restraint-then-fold approach outscoring AlphaFold3’s raw-atom diffusion.

$\text{Mg}^{2+}$  sites and B-factor rigidity deserve emphasis: both are already present inside every mmCIF file the field routinely downloads, are used by no current structure predictor, and were extracted end-to-end in preparing this report.

## 5 Ionic-condition supervision: an audit that rescoped the claim

The headline novelty is accepting ionic conditions as an input. That requires structures labelled with the ionic environment they were solved in, so we audited all 180 sampled structures. The result changed the claim.

Table 12: Where ionic conditions live, and how often they are populated. [measured]

| Method  | $n$ | mmCIF location                                            | usable ionic vector  |
|---------|-----|-----------------------------------------------------------|----------------------|
| Cryo-EM | 117 | <code>_em_buffer_component</code> (structured)            | <b>23.9%</b>         |
| X-ray   | 63  | <code>_exptl_crystal_grow.pdbx_details</code> (free text) | 58.7% mention an ion |

### 5.1 Coverage is roughly one third, split across two storage conventions

Cryo-EM entries record pH for 94% of structures but populate `_em_buffer_component` for only 24.8%. X-ray entries always carry crystallisation details, but as prose—“0.1–0.2 M Arginine-HCl, 0.1M Tris-HCl pH 7.6, 2.6–3.0% PEG-20K”—so extraction is noisy. Combined, about 36% of structures yield any recoverable ionic condition: trainable, but a threefold reduction in usable 3D data for that stage.

### 5.2 The deeper problem: the PDB is survivorship-biased on ionic conditions

Where  $\text{Mg}^{2+}$  concentration *is* recorded, its range is narrow.

Table 13: Recorded buffer concentrations in cryo-EM entries. [measured]

| Ion              | $n$ | p10    | median        | p90     |
|------------------|-----|--------|---------------|---------|
| $\text{Mg}^{2+}$ | 21  | 5.0 mM | <b>7.5 mM</b> | 15.0 mM |
| $\text{K}^+$     | 13  | 30 mM  | 100 mM        | 150 mM  |
| $\text{Na}^+$    | 12  | 100 mM | 150 mM        | 150 mM  |

$\text{Mg}^{2+}$  spans only 5–15 mM. This is not a sampling accident: **nobody deposits a structure of unfolded RNA**. Every PDB entry was solved under conditions chosen *because* the RNA folds there, so the archive contains almost no contrast between folding and non-folding ionic regimes.

**Consequence.** The  $[\text{Mg}^{2+}] \rightarrow$  structure response cannot be learned from the PDB. A model trained only on deposited structures would observe ionic condition as a near-constant and would correctly learn to ignore it.

### 5.3 How the claim is rescoped

The novelty splits into three components with very different evidentiary support.

| Component                                             | Support                                       | Status                |
|-------------------------------------------------------|-----------------------------------------------|-----------------------|
| $B_{\text{elec}}$ Manning/Debye screening             | closed-form physics; needs no training labels | <b>unaffected</b>     |
| $\text{Mg}^{2+}$ site head and inner/outer classifier | 17,428 labelled ions from 180 structures      | <b>well supported</b> |
| Learned <i>response</i> to varying ionic conditions   | requires titration series; the PDB has none   | <b>not supported</b>  |

The first two stand unchanged. The third must be dropped or supplied from data the PDB cannot provide: **RMDB  $\text{Mg}^{2+}$  titration series** (SHAPE/DMS probing across an  $\text{Mg}^{2+}$  ladder on the same sequence) and SAXS  $\text{Mg}^{2+}$  titrations measuring compaction. Training stage 5 as originally specified—ion-conditioned refinement on structures stratified by ionic condition—is **not viable on PDB data alone** and is rewritten to condition on probing titrations, with deposited structures supplying only site-level supervision.

This promotes chemical-probing acquisition from “most valuable missing asset” to **prerequisite for the headline claim**. Note that Ribonanza alone does not supply it; the  $\text{Mg}^{2+}$  titration series in RMDB do.

## 6 Data mapping and the acquisition gap

Six of seven training stages are fully served by the existing catalogue. The exception is chemical probing, and it is the most valuable missing asset. Ribonanza provides 2.1M sequences with DMS and SHAPE measurements—roughly  $315\times$  more supervised examples than the 6,661 unique sequences with 3D structure. It is the only label source that is simultaneously abundant, structure-informative and already public. This is the highest-priority acquisition.

**Corpus weighting.** The eLDORS chunks are source-partitioned rather than homogeneous: 9 of 20 chunks have a median length of exactly 151 nt, indicating unassembled sequencing reads, while chunks 6 and 9 (median 2,193–2,389 nt) are the only substantial source of long-context signal. Uniform sampling in Stage 1 would allow read fragments to dominate pretraining, so per-chunk weighting is required.

## 7 Novelty claims

Table 14: Novelty claims stated precisely, with prior-art status.

| Claim                                                                                                                  | Prior-art status                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ionic conditions as a model input, reaching attention logits through a Manning-screened Coulomb bias                   | No existing RNA structure predictor accepts ionic conditions. The physics term needs no labels; the learned response requires titration data (Section 5). |
| First sparse-MoE architecture for RNA structure                                                                        | AIDO.Protein is MoE for protein; no RNA equivalent exists.                                                                                                |
| Hierarchical coarse-to-fine pair track selecting blocks rather than pairs, justified by measured 1.34% block occupancy | AlphaFold3 and Rhoformer are dense $O(L^2)$ ; flat top- $K$ sparsification is measured here to fail at $\sim 20\%$ recall on long chains.                 |
| Frozen motif KV bank as a retrieval-based geometry prior                                                               | The Motif Atlas is published but wired into no large model.                                                                                               |
| $\text{Mg}^{2+}$ sites and B-factor rigidity as free auxiliary supervision                                             | Present in mmCIF; used by no structure predictor.                                                                                                         |
| Coupled ion-rigidity expert justified by a measured $1.76\sigma$ gradient                                              | Treated separately or not at all elsewhere.                                                                                                               |

## 8 Risks and open questions

1. **Block-detection recall.** Superseded but not eliminated. The flat top- $K$  design was measured at  $\sim 20\%$  recall on long chains and replaced (Section 4.3). The hierarchical track has a 100% recall *ceiling* by construction, but whether a trained model attains it is unproven. Recall must be reported per length bin and per level.
2. **Ionic metadata sparsity—audited, partially confirmed** (Section 5). About 36% of structures yield a recoverable ionic condition. Worse, recorded  $\text{Mg}^{2+}$  spans only 5–15 mM because the PDB is survivorship-biased. The learned ionic response must come from RMDB

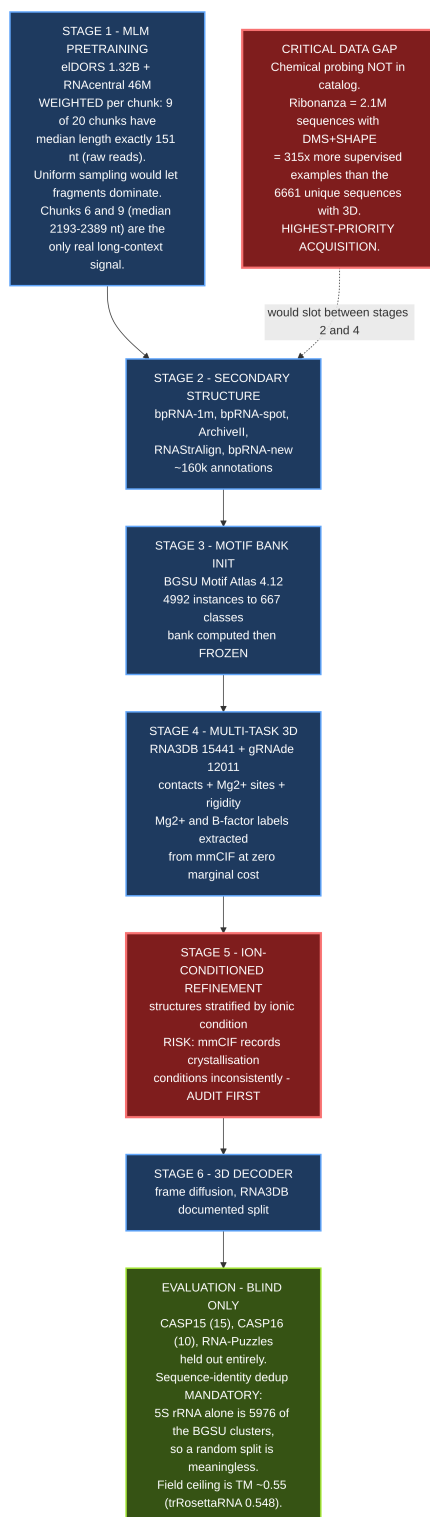


Figure 6: Training curriculum, data sources, and the chemical-probing gap.

titration series, not the PDB. The closed-form physics term and the  $\text{Mg}^{2+}$  site head are unaffected.

3. **Crystallographic ions are not the solution ensemble.** The explicit head learns site binding, not the diffuse atmosphere. These must not be conflated.
4. **B-factor is a noisy rigidity proxy.** Excluding cryo-EM left only 31 of 137 parsed structures usable, which materially shrinks the rigidity label set.
5. **Association, not isolated causation.**  $\text{Mg}^{2+}$  proximity and packing density both track folded cores.
6. **Infrastructure.** NucleicBERT used 192 A100s densely. MoE improves FLOP efficiency but adds routing and communication complexity.
7. **Evaluation honesty.** The field ceiling is  $\text{TM} \approx 0.55$ . Any improvement claim must be made on blind sets—CASP15/16 and RNA-Puzzles—never on rRNA-saturated random splits. Sequence-identity deduplication is mandatory.

## 9 Status

This is a design specification. No model has been trained and no predictive performance is claimed. What has been established is the evidentiary basis: the contact-scaling law, the ion-rigidity coupling, the ion inventory and coordination chemistry, and the GNRA negative result, all measured on 180 non-redundant structures with reproducible scripts. The next phase is to implement the sparse pair track and validate proposal recall before any large-scale training is attempted.

## A Reproducing the measurements

| Script                                                      | Output                                 |
|-------------------------------------------------------------|----------------------------------------|
| <code>scripts/sampling/fetch_samples.py</code>              | 180 mmCIF, 12 Rfam seeds               |
| <code>scripts/sampling/analyze_ions_motifs.py</code>        | <code>ion_summary.json</code>          |
| <code>scripts/sampling/analyze_rigidity.py</code>           | <code>rigidity_summary.json</code>     |
| <code>scripts/sampling/analyze_contact_sparsity.py</code>   | <code>contact_sparsity.json</code>     |
| <code>scripts/sampling/validate_proposal_recall.py</code>   | <code>proposal_recall.json</code>      |
| <code>scripts/sampling/analyze_contact_separation.py</code> | <code>contact_separation.json</code>   |
| <code>scripts/sampling/analyze_block_sparsity.py</code>     | <code>block_sparsity.json</code>       |
| <code>scripts/sampling/audit_ionic_metadata.py</code>       | <code>ionic_metadata_audit.json</code> |
| <code>scripts/sampling/audit_em_buffers.py</code>           | <code>em_buffer_audit.json</code>      |

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